Internal Fidelity and Real-Space Locality of Reduced Wannier Hamiltonians: A Rank-10 Silicon Case Study

Project Koios

Development technical report, version 2

Abstract

We study how real-space reduction changes a fixed, rank-10 Wannier Hamiltonian derived from a scalar-relativistic silicon calculation. The purpose is not to validate the Hamiltonian as a quantitatively accurate model of silicon. Instead, it is to determine which internal fidelity properties survive changes of gauge, literal removal of complete Wigner–Seitz shells, and bounded refitting of retained shells.

A constructed momentum-dependent unitary leaves the complete represented spectrum invariant to 3.20×10^{-14} eV while changing restricted alignment residuals and Fourier-block locality. Literal nested shell deletion produces an almost one-dimensional spectral–operator loss family and requires 371 of 383 representatives for the tightest proper reduction. Bounded shell refitting creates genuine same-support spectral–operator tradeoffs and reduces the observed support at several post-hoc error pairs. However, errors measured on the 408-point fitting inventory understate exact integrated operator error and independent dense-mesh spectral error. Band-edge quantities are still more demanding: moderate supports substantially alter conduction masses, split the nearly degenerate valence manifold, and amplify differences among symmetry-equivalent high-symmetry points.

The results show that spectral agreement, operator agreement, real-space locality, and band-edge fidelity are distinct properties. Near-full support is required to preserve all of them for this represented source. Because the source itself fails separate DFT-parent gap and electron-mass targets, every conclusion is restricted to internal Wannier-model fidelity.

Evidence status. All numerical results in this report are checksum-bound, producer-checked development outputs. They have not been independently reconstructed. No Quantum ESPRESSO or Wannier90 process was invoked for the analyses reported here, and no frozen 402-point confirmation values were consumed.

1 Introduction

Wannier Hamiltonians provide a finite-dimensional representation in which a sampled electronic structure can be interpolated and expressed through real-space matrix blocks [1, 2, 3, 4]. This representation is attractive for reduced models because the Hamiltonian can be written as a Fourier series and approximated by a smaller set of real-space terms. The apparent simplicity of that procedure hides several independent questions.

First, the band energies do not uniquely determine the matrix representation. A momentum-dependent unitary transformation can preserve all eigenvalues while changing the eigenvectors and the Fourier coefficients. Locality is therefore a property of a represented Hamiltonian in a chosen

gauge, not a property of the spectrum alone. Second, removing real-space terms changes both the matrix field and its eigenvalues, but these two errors need not rank candidate models in the same way. Third, a small global spectral error does not guarantee preservation of derivatives, extrema, degeneracies, or effective masses.

This report organizes one development campaign around those distinctions. The source is the authenticated $C6$-B20-W21 rank-10 Wannier Hamiltonian for bulk silicon. We ask four questions:

1. Can a complete represented spectrum remain unchanged while real-space locality and restricted frame alignment change?
2. How much native Wigner–Seitz support is required when shells are literally deleted?
3. Does bounded refitting of retained shells create useful spectral–operator tradeoffs?
4. Do models selected from the 408-point development inventory remain faithful on dense reciprocal meshes and at conduction and valence band edges?

The answer to the first question is yes, by construction. The second and third answers are more restrictive: literal deletion retains most of the native support at tight errors, while refitting helps mainly at moderate post-hoc tolerances. The fourth answer is the central practical result. Global fitting errors systematically understate broader interpolation and band-edge consequences, and a nearly full-support model is needed to preserve the source across all evaluated channels.

2 Background

2.1 Wannier representation and gauge freedom

For a rank-N periodic Hamiltonian, the represented matrix field is

$$H(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} T(\mathbf{R}), \tag{1}$$

where $T(\mathbf{R})$ is an $N \times N$ hopping block. A momentum-dependent unitary $U(\mathbf{k})$ defines another representation

$$\widetilde{H}(\mathbf{k}) = U(\mathbf{k})^{\dagger} H(\mathbf{k}) U(\mathbf{k}). \tag{2}$$

The two fields have the same complete spectrum at each momentum, but their Fourier blocks generally differ. Consequently, a statement that a Hamiltonian is "local" requires a declared basis, gauge, lattice convention, and Fourier route.

This report uses the word *gauge* only for unitary transformations inside the fixed rank-10 represented space. The constructed attack is not presented as a typical Wannier90 gauge change. It is an adversarial control that demonstrates what the spectrum alone cannot determine.

2.2 Matrix fidelity and spectral fidelity

Let $H_{\mathrm{W}}(\mathbf{k})$ be the full source and $H_{\mathrm{r}}(\mathbf{k})$ a reduced model in the same Wannier basis. On the uniform 408-point development inventory, the normalized ordered-spectrum and operator RMS losses are

$$\tau_1 = \left[\frac{1}{4080} \sum_{\mathbf{k},n} \left(\frac{\epsilon^{\mathrm{r}}_{n\mathbf{k}} - \epsilon^{\mathrm{W}}_{n\mathbf{k}}}{1\,\mathrm{eV}} \right)^2 \right]^{1/2}, \tag{3}$$

$$\tau_2 = \left[\frac{1}{4080} \sum_{\mathbf{k}} \frac{\| H_{\mathrm{r}}(\mathbf{k}) - H_{\mathrm{W}}(\mathbf{k}) \|_{\mathrm{F}}^2}{(1\,\mathrm{eV})^2} \right]^{1/2}. \tag{4}$$

The 1 eV factor is a reporting normalization, not an acceptance threshold. The operator loss uses identity alignment because all shell models remain in the same frozen Wannier basis. A pointwise fitted alignment would remove part of the model error that the reduction is intended to measure.

Spectral error is bounded by matrix error in standard perturbation results, but the converse does not hold. Eigenvalues can agree while the matrix representation, frame, and real-space coefficients differ. The two losses are therefore retained as separate coordinates rather than scalarized into one acceptance score.

2.3 Why band-edge fidelity is separate

A global RMS average gives equal formal status to all sampled bands and points. A band edge instead depends on a local extremum and its derivatives. For a nondegenerate band, the inverse effective-mass tensor is proportional to the Cartesian Hessian,

$$(m^{-1})_{ij} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k_i \partial k_j}. \tag{5}$$

Small energy errors can produce much larger curvature errors after differentiation.

The valence edge introduces an additional difficulty. Near a degenerate manifold, independently ordered eigenvalues need not define analytic branches. A basis- and ordering-independent analysis must operate on the coupled subspace, as in multiband $\mathbf{k} \cdot \mathbf{p}$ reasoning [5]. Because the present source is non-SOC, we do not assign physical heavy-hole, light-hole, or split-off labels. Instead, we fit a matrix-valued quadratic Hamiltonian for the three-band valence cluster.

3 Methodology

3.1 Represented source and scope

The source is the full, untruncated $C6$-B20-W21 Hamiltonian with rank 10 and 383 unique effective native Wigner–Seitz representatives. It was constructed from a Gamma-origin $6 \times 6 \times 6$ Wannier mesh with 20 parent bands and a rank-10 target space. Structural checks established pair closure, real-space Hermiticity, source reconstruction, and reciprocal periodicity.

The source is suitable for an internal reduction study but not for a claim of accurate silicon band-edge physics. In a separate revealed 408-point DFT-parent comparison it missed the declared indirect-gap and effective-mass targets. We therefore treat H_{W} as the represented reference and never reinterpret source-relative agreement as agreement with DFT or experiment.

Two Fourier routes appear in the campaign and are kept distinct:

- the native Wannier90 HR/Wigner–Seitz representation used for shell reduction;

- a centered finite-cyclic transform of sampled matrix fields, followed by explicit cyclic-to-ordinary conversion, used for the gauge experiment.

Native and finite-cyclic blocks are not silently substituted for one another.

3.2 Constructed gauge attack

The attack is a frozen rank-10, depth-2 alternating Givens circuit. Its harmonics use the three Cartesian-axis bonds and an axis-maximum normalization with coefficient $\pi/8$. Four comparison channels are retained:

1. identity coordinates;
2. one momentum-independent global Procrustes unitary;
3. pointwise polar alignment;
4. the known generating inverse, used only as an oracle control.

The reference and attacked fields are sampled on the centered 6^3 construction inventory, transformed to finite-cyclic blocks, and converted to 125 ordinary representatives with components in $\{-2,-1,0,1,2\}$. Complete physical-radius supports are then evaluated on the ordered 408-point development inventory. The study separates complete-field gauge equivalence, conversion/interpolation error, deliberate support truncation, and total finite-model error.

3.3 Literal native-shell reductions

The native representatives are grouped into 18 complete physical-radius shells. H-shell-00 is the onsite-only control; H-shell-17 is the full 383-representative source. The 17 proper reductions H-shell-00 through H-shell-16 are evaluated without fitting. For a post-hoc pair (τ_1, τ_2), the witness surface reports the smallest proper support satisfying both inequalities. Full support is retained as a zero-loss control but excluded from reduced-family feasibility.

Because this family is nested and has only one support coordinate, both losses are driven by the same omitted matrix field. A nearly one-dimensional sequence of requirement points is therefore expected and should not be interpreted as a general theorem that spectral and operator losses are interchangeable.

3.4 Refitted shell models

To test whether the literal result is caused by the absence of adjustable parameters, each retained complete shell is assigned one real scale,

$$T_{\boldsymbol{\alpha}}(\mathbf{R}) = \alpha_s T_{\mathrm{W}}(\mathbf{R}), \qquad 0 \le \alpha_s \le 2. \tag{6}$$

The same scale applies to every representative in shell s, including the Hermitian partner at $-\mathbf{R}$. Blocks outside the support remain zero. This model preserves the fixed rank-10 Wannier basis and the within-shell block pattern; it is not a Slater–Koster model and carries no atomic-orbital semantics.

For each proper support, 17 spectral–operator scalarization weights and three deterministic starts are optimized with bounded L-BFGS-B and analytic gradients. The search produces constructive witnesses only. A failed or absent optimization record is not an infeasibility certificate.

3.5 Frozen fidelity ladder

Three converged witnesses are frozen before broader grids are inspected: H-shell-05 with 79 representatives, H-shell-10 with 201, and H-shell-16 with 371. No parameters are refitted later.

The exact normalized-Brillouin-zone operator RMS is obtained from the unique Fourier coefficients by Parseval's identity. Spectral errors are then evaluated on independent Gamma-origin 12^3, 20^3, and 32^3 meshes. Finally, the six conduction valleys are located from sector searches and local optimization, and source-relative gaps, positions, Hessians, and masses are calculated with identical algorithms.

3.6 Comprehensive band-edge analysis

For each conduction valley, Cartesian Hessians are evaluated with central-difference steps of 0.001, 0.002, 0.005, and 0.01 Å^{-1}. The 0.002 Å^{-1} result is primary. In addition to full tensors and principal masses, we report the one-valley density-of-states mass convention, the one-valley conductivity-mass convention, a six-valley DOS-mass convention, and an equal-population average inverse-mass convention. These are curvature summaries, not transport or finite-temperature carrier calculations.

At Gamma, bands 1–3 in zero-based ordering define the upper valence cluster. We first retain its splitting, projector distance, and principal angles. Ordered directional fits along [100], [110], and [111] are used only to demonstrate why individual branch masses are unstable.

The primary valence analysis polar-transports the displaced three-band eigenspace to its Gamma cluster and fits the even represented field to

$$H_{\text{eff}}(\mathbf{q}) = H_{\text{eff}}(0) + q_x^2 B_{xx} + q_y^2 B_{yy} + q_z^2 B_{zz} \tag{7}$$

$$+ 2q_x q_y B_{xy} + 2q_x q_z B_{xz} + 2q_y q_z B_{yz}. \tag{8}$$

Thirteen unoriented rays and four positive radii per ray are used at each of three fit windows. Directional curvature channels are the eigenvalues of twice the contracted quadratic matrix. Candidate coefficient fields are polar-aligned to the source Gamma cluster before their joint Frobenius error is calculated.

4 Results

4.1 Spectrum-preserving gauge changes redistribute locality

The maximum complete-spectrum difference between the source and attacked represented fields is 3.197×10^{-14} eV. The known-inverse and pointwise-polar controls recover the frame to near machine precision, with maximum defects of approximately 5.06×10^{-16} and 1.29×10^{-15}. Identity and one-global-unitary alignment leave order-one maximum frame defects of approximately 1.021 and 1.020.

Despite complete spectral invariance, the non-onsite finite-block power changes from 20.7574 to 21.5057 eV^2 per retained state. The shell-resolved and cumulative power distributions change accordingly. This is a direct counterexample to the idea that the complete spectrum fixes real-space locality.

Finite-support requirements also change. The reference and attacked support-error curves have different non-dominated supports even though their complete represented spectra agree. The result concerns the constructed attack, not the distribution of ordinary Wannier90 gauges.

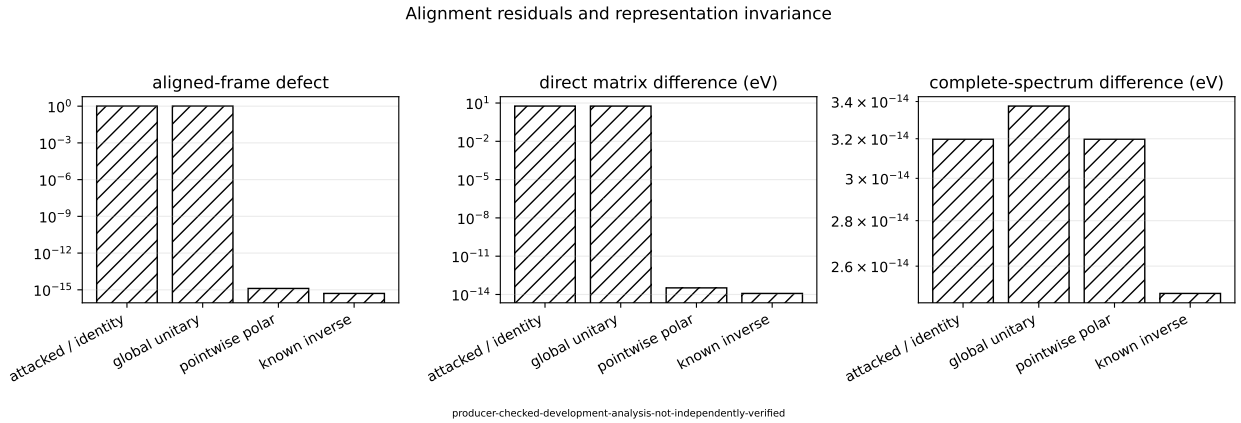


Figure 1: Complete spectral invariance and alignment diagnostics for the frozen gauge attack. Pointwise polar and known-inverse channels are oracle controls; identity and one global unitary cannot remove the momentum-dependent attack.

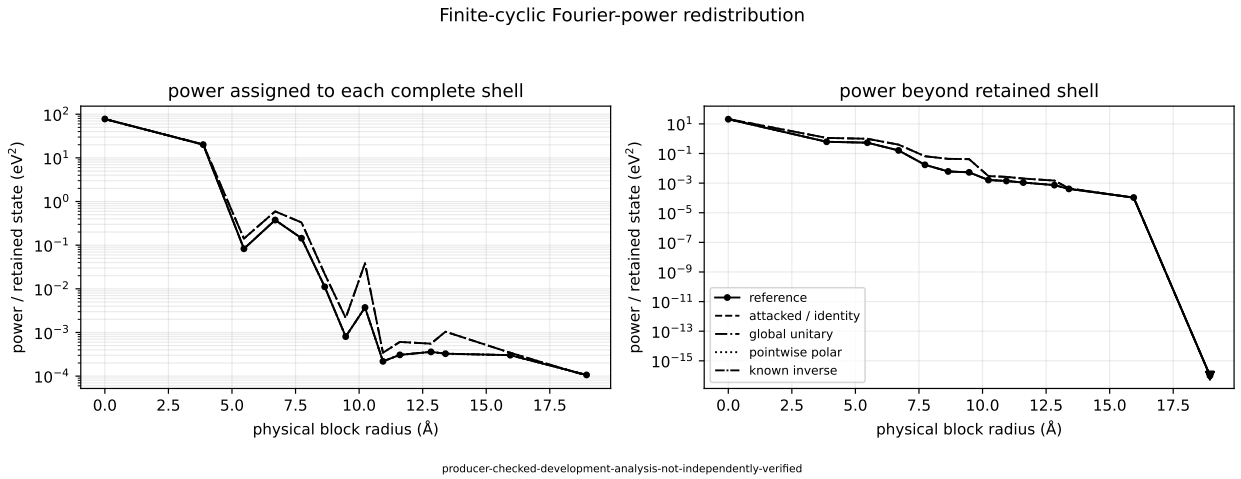


Figure 2: Redistribution of finite Fourier-block power under the spectrum-preserving attack. The cyclic-to-ordinary removed power is shown separately and is not counted as deliberate support truncation.

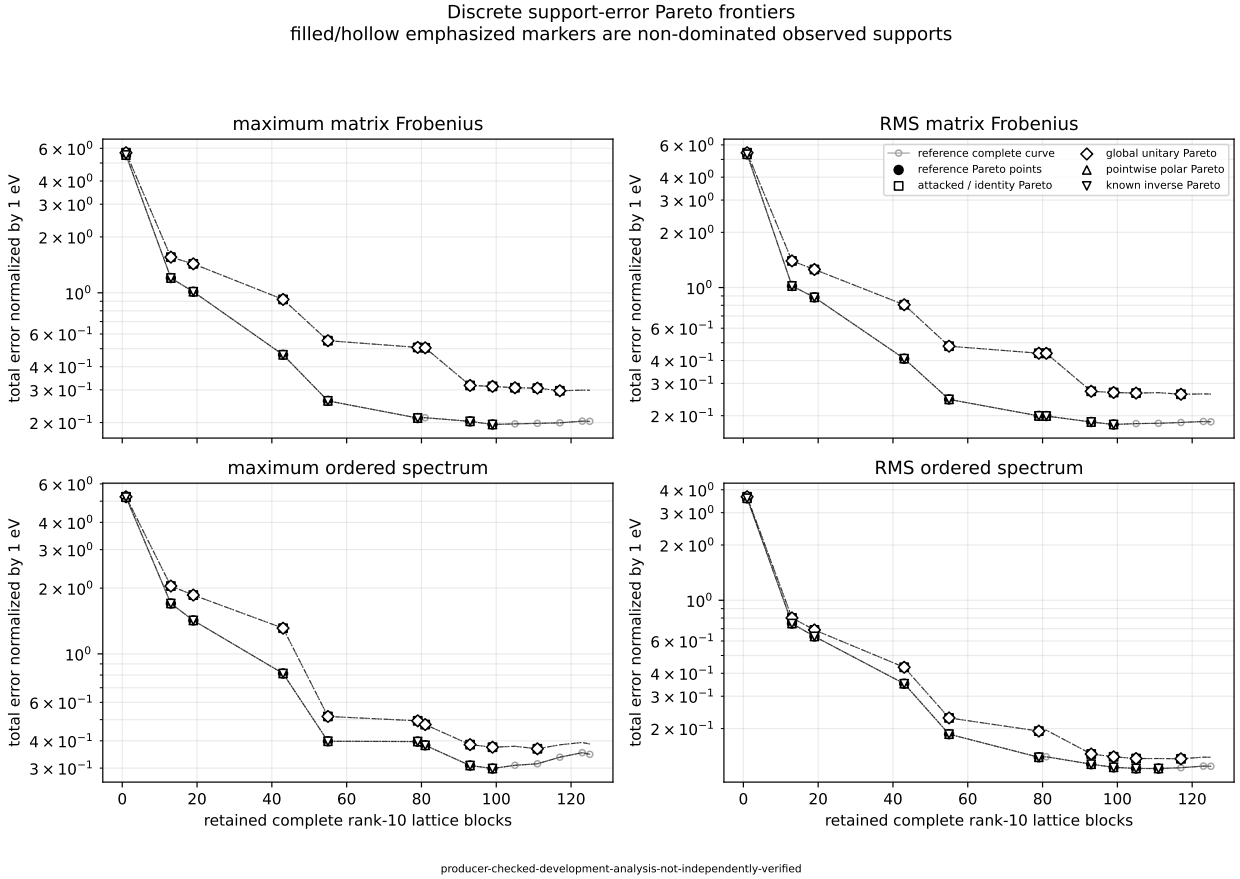


Figure 3: Discrete complete-support error frontiers for reference and attacked representations. Markers identify observed non-dominated supports; connecting segments are visual guides and do not interpolate unobserved support choices.

4.2 Literal shell deletion retains most of the source at tight errors

Among all proper literal reductions, H-shell-16 minimizes both primary RMS losses while retaining 371 of 383 representatives:

$$(\tau_1, \tau_2) = (4.4636 \times 10^{-4}, 1.5173 \times 10^{-3}). \tag{9}$$

Thus the literal nested family does not display a global spectral–operator conflict at its tightest proper reduction. Smaller supports show local nonmonotonicity, but the requirement points remain close to a line because one nested support coordinate controls both errors.

The post-hoc pair $(0.1, 0.15)$ first admits 135 representatives; $(0.01, 0.02)$ requires 313; and $(0.001, 0.002)$ requires 371. No proper literal reduction satisfies $(0.0004, 0.001)$. These coordinates summarize the observed family and are not prospective acceptance thresholds.

4.3 Shell refitting creates genuine two-loss tradeoffs

Bounded shell scaling changes the geometry of the candidate family. A fixed support can now trace a spectral–operator tradeoff arc rather than contribute only one deletion point. Across 444 converged witnesses, refitting reduces the minimum observed support at several illustrative pairs (Table 1).

Table 1: Observed minimum-support literal and refitted witnesses. Threshold pairs are post-hoc descriptive coordinates, not acceptance limits.

(τ_1, τ_2)	Literal representatives	Refitted representatives
$(1.0, 1.0)$	19	13
$(0.1, 0.15)$	135	79
$(0.05, 0.1)$	177	135
$(0.02, 0.04)$	225	201
$(0.01, 0.02)$	313	313
$(0.001, 0.002)$	371	371

Within H-shell-16, an operator-oriented witness attains approximately $(4.04 \times 10^{-4}, 1.394 \times 10^{-3})$, while a spectral-oriented witness attains $(2.98 \times 10^{-4}, 1.745 \times 10^{-3})$. This is a genuine same-support tradeoff, although bounded multistart optimization does not establish global optimality.

4.4 The 408-point inventory understates broader error

The frozen fidelity ladder gives the central quantitative comparison. For every model, the fitted-408 spectral and operator losses are smaller than their broader counterparts (Table 2).

Table 2: Fitted-inventory and broader frozen-model RMS errors.

	Spectral RMS (eV)			Operator RMS (eV)		
Model	408	32^3	Ratio	408	Parseval	Ratio
H-shell-05	0.07242	0.18196	2.51	0.12814	0.26004	2.03
H-shell-10	0.01320	0.02785	2.11	0.03229	0.05956	1.84
H-shell-16	0.000404	0.000657	1.62	0.001394	0.001747	1.25

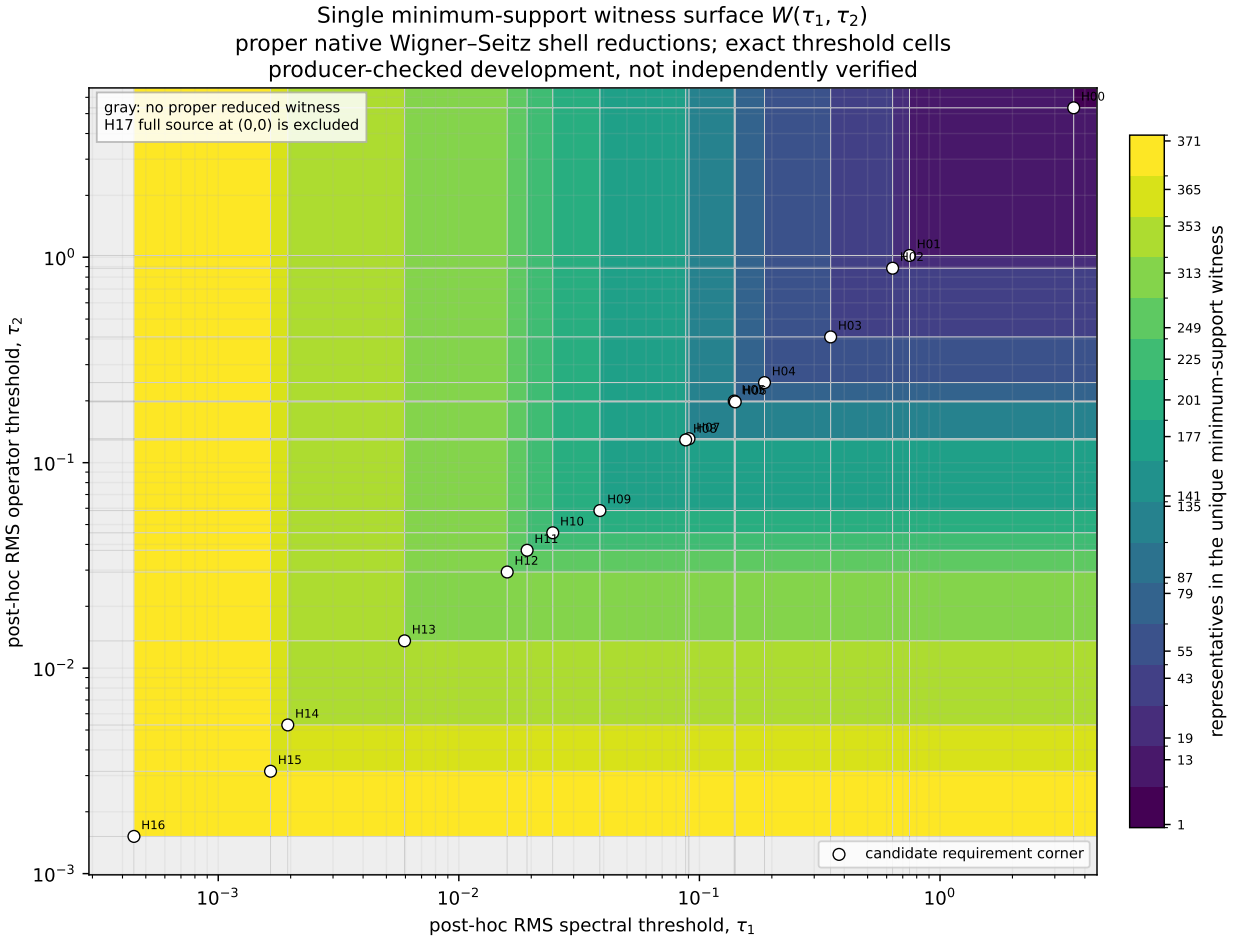


Figure 4: Minimum-support witness surface for literal complete-shell deletion. The full source is excluded from reduced-family feasibility. Gray cells contain no proper literal witness at the displayed post-hoc pair.

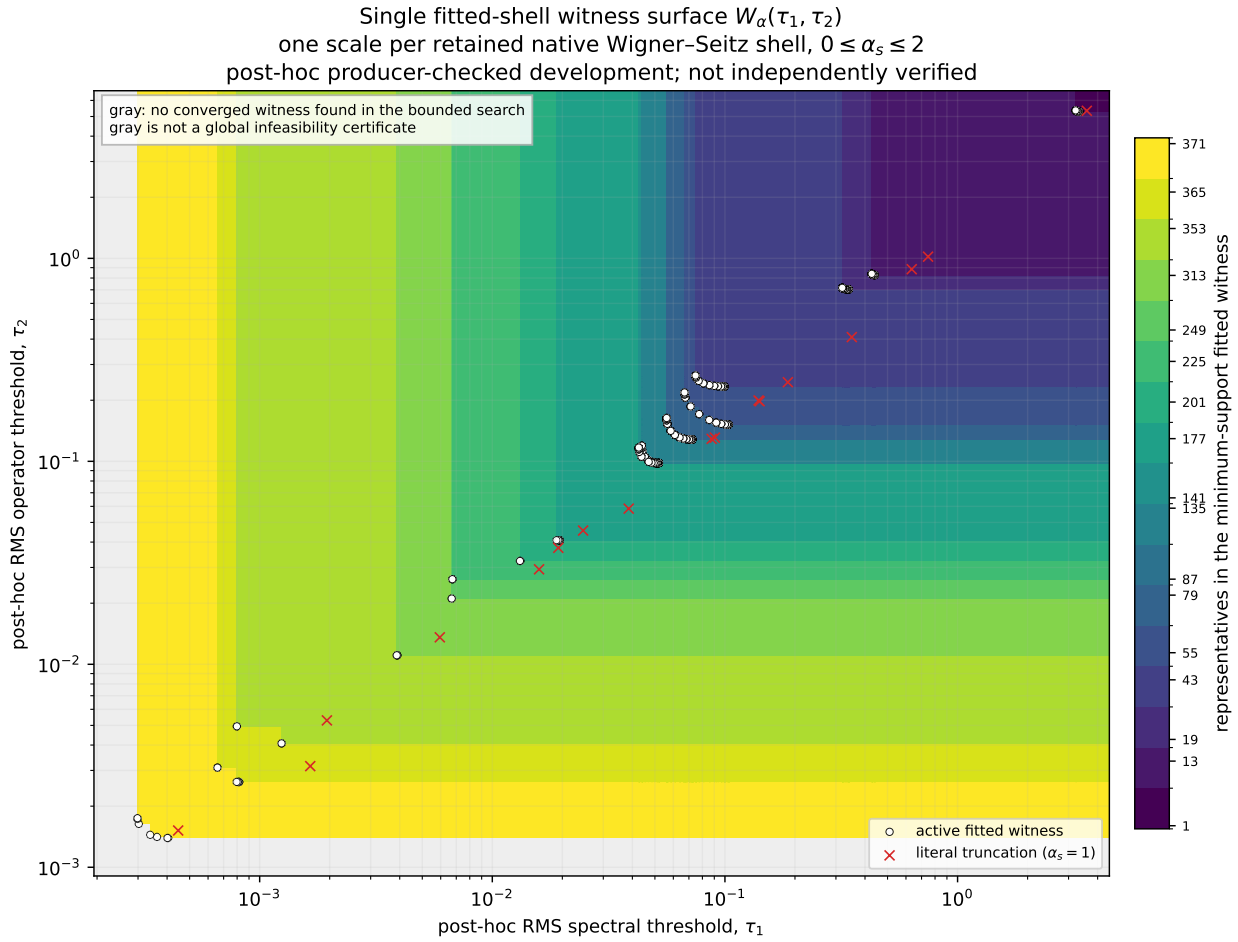


Figure 5: Minimum-support surface over converged refitted shell witnesses. Gray means that the bounded search found no converged witness, not that the pair is mathematically infeasible.

The sampled operator RMS agrees with the exact Parseval value to at most 5.56×10^{-17} eV. Spectral RMS changes between the 12^3 and 32^3 meshes remain below about 1.4%, so the dense values are stable over the tested mesh sequence. The discrepancy is therefore not a late-grid fluctuation; it is a difference between fitting-inventory and broader-domain evaluation.

4.5 Band-edge features are more demanding than global RMS

H-shell-05 changes the source-relative indirect gap by -65.8 meV, displaces a conduction valley by as much as 0.0948 Å^{-1}, and changes at least one principal mass by 67.7%. H-shell-10 has a 32^3 all-band spectral RMS below 0.028 eV but still changes a principal mass by 53.2%. H-shell-16 changes the gap by only -0.503 meV, the maximum valley position by 0.00276 Å^{-1}, and the maximum principal mass by 2.54%.

The expanded tensor analysis leads to the same ordering. The source six-valley DOS-mass convention is $1.4764\, m_e$, and its equal-population conductivity-mass convention is $0.25063\, m_e$. H-shell-05 changes them by 16.9% and 20.4%; H-shell-10 by 22.9% and 6.90%; and H-shell-16 by 0.333% and 0.0216%. The maximum mass variation across four finite-difference steps is below 0.172% for every model, so stencil choice does not explain the large moderate-support errors.

4.6 Moderate reduction damages the degenerate valence edge

The full source has a three-band Gamma splitting of only 0.02495 meV. H-shell-05 and H-shell-10 increase this to 102.2 and 54.8 meV. H-shell-16 reduces the error substantially but still yields 2.50 meV. The corresponding maximum source-cluster principal angles are 1.856, 0.925, and 0.0420 degrees. The moderate models therefore preserve a recognizable three-dimensional subspace while strongly changing the Hamiltonian inside it.

Independently ordered source branches do not yield stable hole masses: the upper branch is not hole-like for every tested window in any of the three directions, and finite ordered-branch masses vary by up to about 75.5%. The primary multiband fit avoids that assignment. At the 0.01 Å^{-1} window, its source curvature-channel masses are listed in Table 3.

Table 3: Full-source non-SOC valence curvature-channel masses in electron-mass units.

Direction	Channel 1	Channel 2	Channel 3
[100]	0.1411	0.2771	0.3015
[110]	0.09690	0.3119	2.923
[111]	0.08770	0.6971	0.7722

These channels belong to a scalar-relativistic three-band quadratic operator. They are not physical SOC heavy-hole, light-hole, or split-off masses. After cluster alignment, H-shell-05, H-shell-10, and H-shell-16 have quadratic-coefficient errors of 9.54%, 5.88%, and 0.279%, and maximum curvature-channel mass changes of 38.9%, 19.7%, and 4.53%.

A related symmetry diagnostic reaches the same conclusion. The full source's direct-gap spread among equivalent X points is 0.00366 meV and among L points is 0.0239 meV. H-shell-05 increases those spreads to 150.1 and 68.7 meV; H-shell-10 to 36.7 and 63.9 meV; and H-shell-16 to 0.427 and 2.97 meV. Moderate shell refitting can therefore amplify small residual symmetry differences through cancellations among retained blocks.

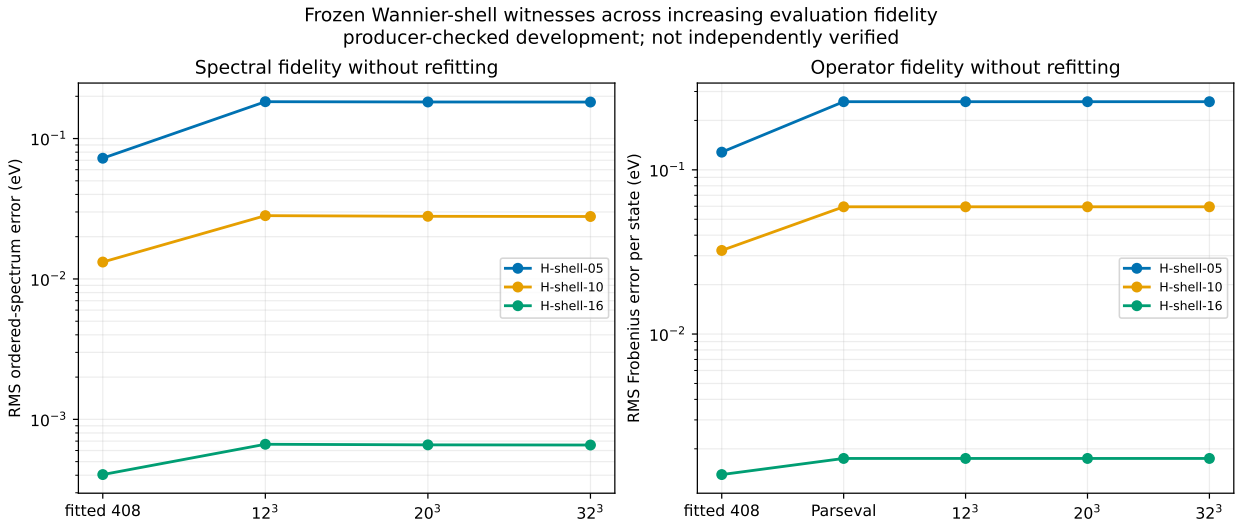


Figure 6: Frozen-model fidelity from the fitted inventory to exact and dense evaluations. Supports and shell scales remain unchanged at every evaluation level.

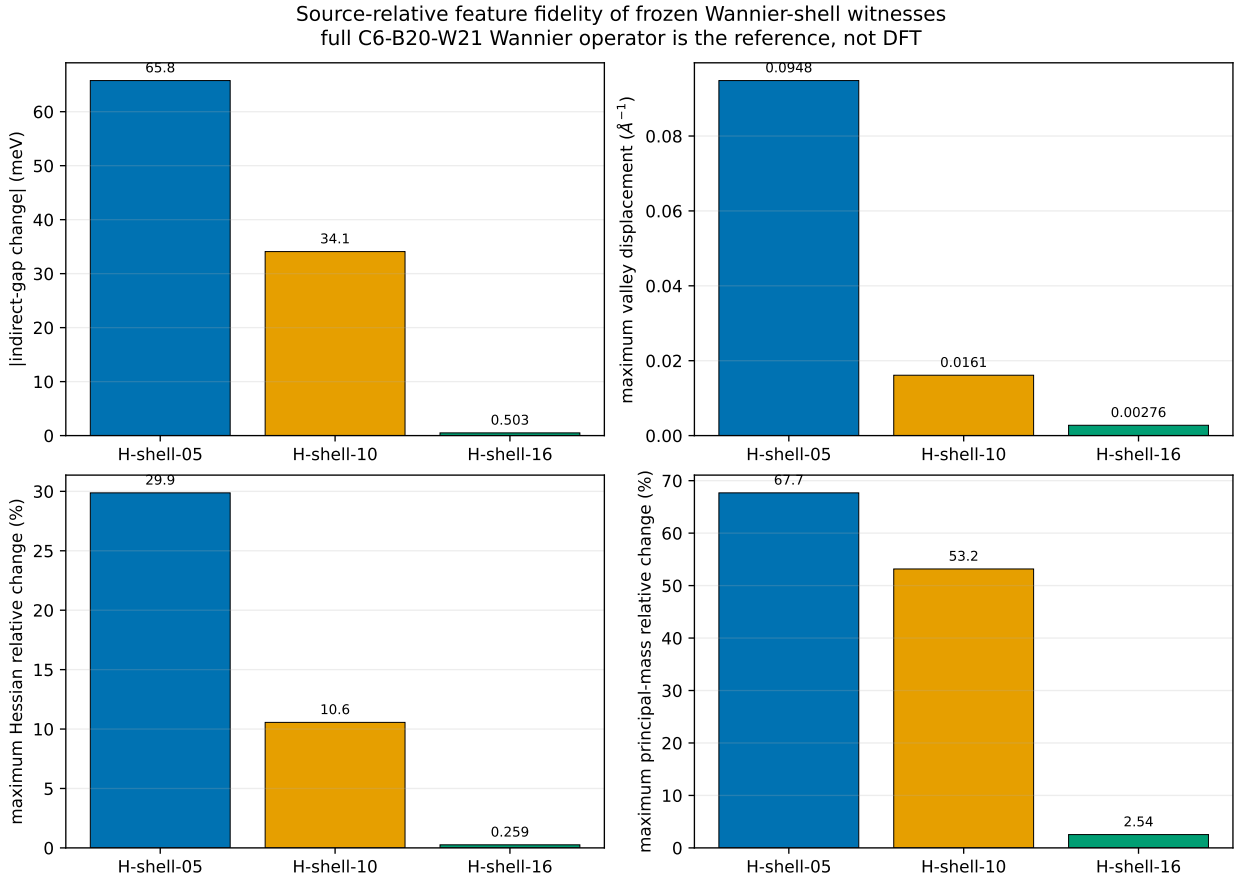


Figure 7: Source-relative conduction-edge consequences of the frozen shell models. The reference is the full Wannier source, not DFT or experiment.

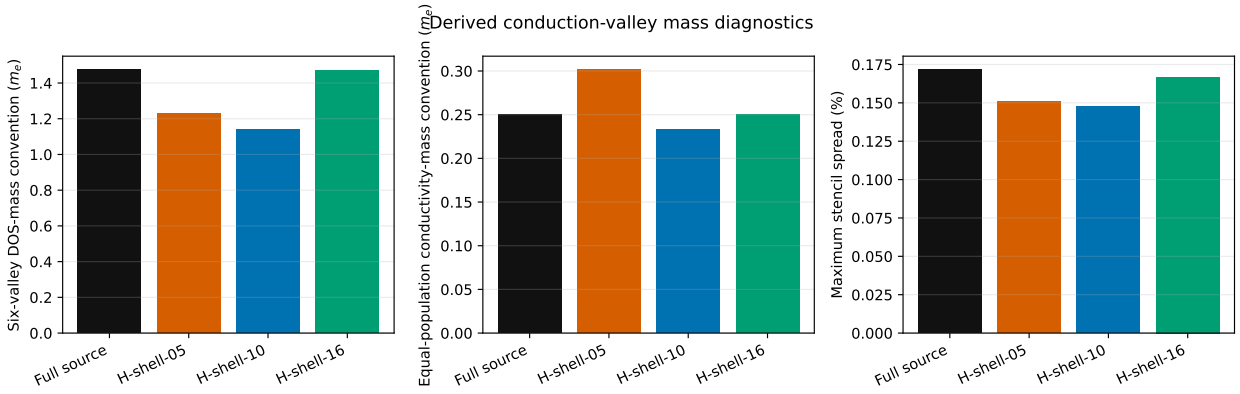


Figure 8: Derived conduction-valley mass conventions and finite-difference sensitivity. The DOS and conductivity quantities are curvature conventions, not carrier or transport calculations.

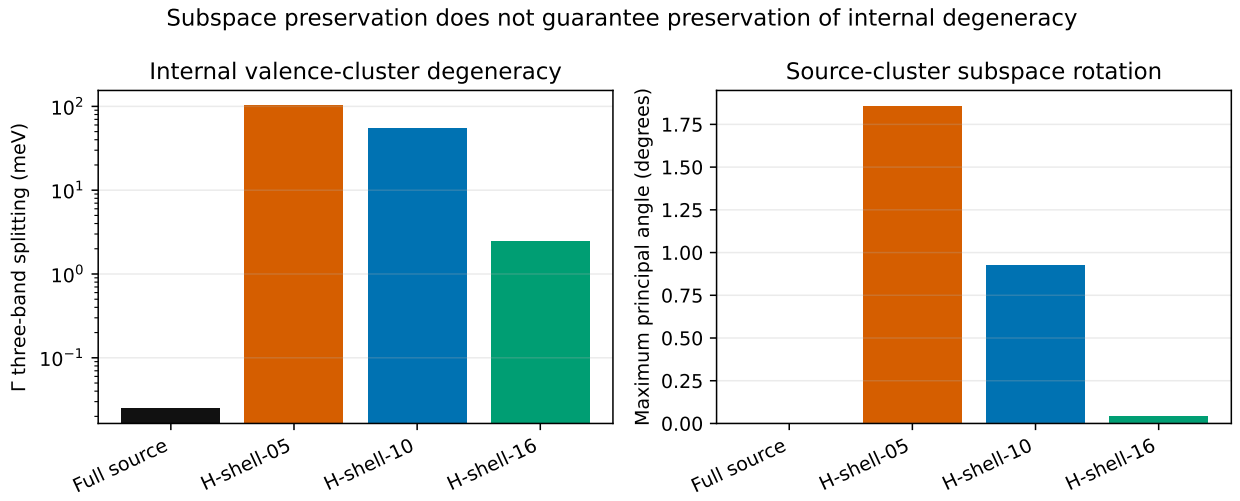


Figure 9: Gamma valence-cluster splitting and subspace rotation. Subspace agreement and preservation of internal degeneracy are separate conditions.

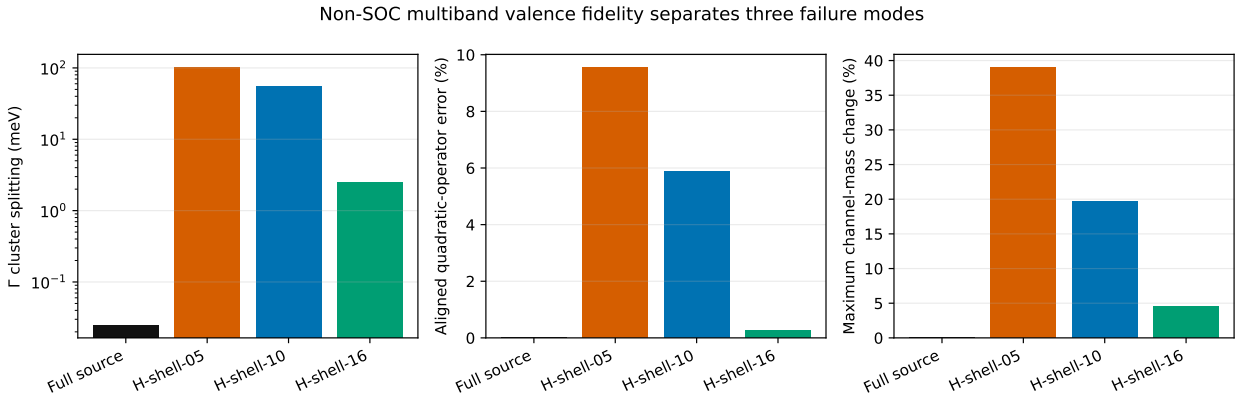


Figure 10: Summary of non-SOC multiband valence fidelity. Gamma splitting, aligned quadratic-operator error, and maximum curvature-channel change expose distinct failure modes.

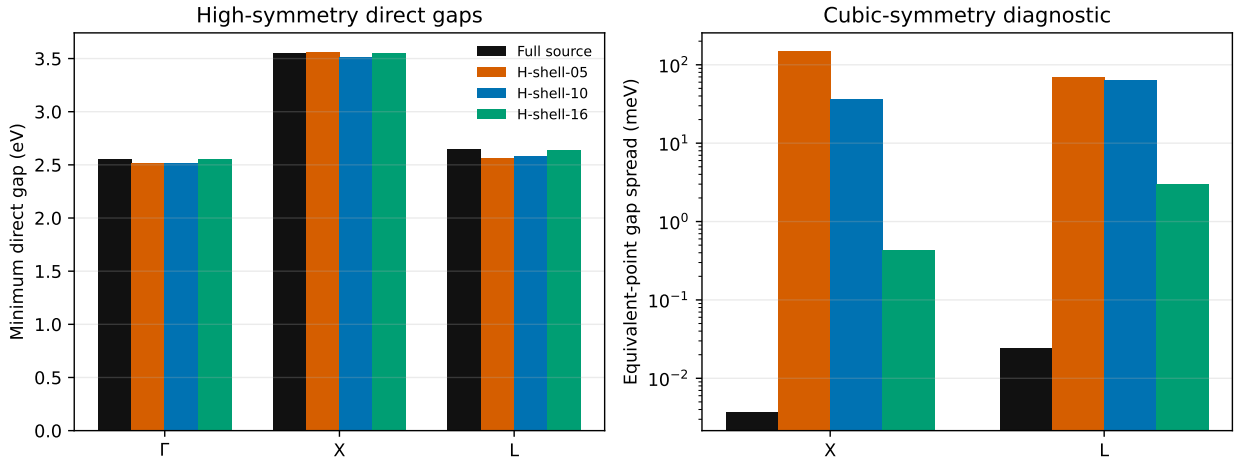


Figure 11: Direct-gap and equivalent-point symmetry diagnostics. The displayed spreads characterize the represented models and are not an independent crystal-symmetry validation.

5 Discussion

5.1 Locality is representation-dependent

The gauge experiment supplies the cleanest conceptual result. The complete spectrum is unchanged to roundoff, yet block power and finite-support requirements change. Thus a band structure cannot determine a unique hopping-locality profile. Any comparison of "short-range" Hamiltonians must identify the represented state space and the allowed gauge transformations before locality is interpreted.

This does not imply that locality is arbitrary or useless. A fixed construction, gauge, and basis produce a well-defined optimization problem. It does imply that locality claims cannot be elevated from spectral data alone, and that a constructed gauge counterexample should not be confused with ordinary variation between independently localized Wannier calculations.

5.2 Refitting helps, but support remains the dominant resource

Literal shell deletion is too restrictive to reveal a strong global two-loss conflict: one nested coordinate controls both matrix and spectral losses. Adding bounded shell scales exposes genuine tradeoff arcs and reduces support at moderate post-hoc error pairs. This shows that some error from deletion can be compensated by renormalizing retained shells.

The compensation has limits. At tight errors the observed minimum support remains 371 representatives, close to the 383-representative source. Moreover, the bounded search is constructive rather than exhaustive. The correct conclusion is that refitting produces useful witnesses, not that the displayed surface is a global optimum or that gray cells are infeasible.

5.3 A fitting inventory is not an interpolation guarantee

Every frozen model performs worse on the exact integrated or independent dense evaluation than on the 408-point inventory used to characterize fitted witnesses. The discrepancy is largest at moderate support and decreases near full support. This is the expected role of a broader evaluation: it reveals behavior between fitted points without changing the model.

No held-out set is required to make the finite statement that a model attains a given loss on the 408 retained points. A held-out or all-$\mathbf{k}$ argument is required only when the claim expands to interpolation or Brillouin-zone behavior. Parseval's identity provides the exact integrated operator statement here, while dense meshes provide the spectral interpolation check.

5.4 Band edges should be treated as first-class objectives

The conduction results show why all-band RMS is an incomplete reduction target. H-shell-10 appears moderately accurate in the dense global spectral metric, yet changes a principal electron mass by more than 50%. Effective masses depend on second derivatives and therefore amplify local shape errors that contribute little to a global average.

The valence result is stronger. Independent band Hessians are not merely noisy; at the near-degenerate source edge they are not a stable mathematical object. The appropriate quantity is a matrix-valued multiband quadratic Hamiltonian. Moderate reductions can make individual branches look numerically stable by incorrectly splitting the manifold. Stability after symmetry breaking is not fidelity.

For this non-SOC source, the multiband operator is the strongest justified hole-sector result. Physical silicon heavy-hole, light-hole, split-off, or acceptor claims would require an explicitly spinor,

SOC-enabled source and a separately qualified comparison. SOC is not required for the internal compression conclusion reached here.

5.5 What the study establishes

The retained calculations support the following bounded conclusions:

1. Complete spectral equivalence does not determine real-space block locality.
2. Operator and spectral losses are distinct and should remain separate reduction coordinates.
3. Shell refitting can reduce support at moderate post-hoc losses, but tight internal fidelity still requires near-full native support.
4. Errors on the fitted 408-point inventory understate broader interpolation error.
5. Global spectral RMS does not control conduction masses, valence degeneracy, or multiband curvature.
6. Among the frozen witnesses, H-shell-16 is consistently closest to the source, while H-shell-05 and H-shell-10 are not adequate band-edge reductions.

5.6 Limitations and next steps

The principal limitation is the represented source itself. It fails separate DFT-parent gap and electron-mass criteria, so source-relative preservation does not validate silicon physics. The source is also scalar-relativistic and non-SOC. The analyses are producer-checked but not independently reconstructed, and all tolerance surfaces are post hoc rather than prospectively accepted.

A scientifically stronger continuation would keep the current models frozen and add, in order:

1. independent reconstruction of the retained analyses;
2. a clearly separated overlay against the existing revealed 408-point DFT parent, without refitting to that overlay;
3. a qualified SOC DFT/Wannier branch if physical hole or acceptor properties are required; and
4. prospective reduction thresholds declared before inspecting any future confirmation set.

None of these steps changes the present internal conclusion: for the tested source and model family, near-full support is required to preserve matrix, spectral, symmetry, and band-edge behavior simultaneously.

5.7 Reproducibility

Canonical requests, authenticated source identities, producer programs, machine-readable results, figure manifests, and execution records are retained under `calculations/Si/bulk/analysis/`. The gauge, literal-shell, refitted-shell, fidelity-ladder, and comprehensive-band-edge analyses each retain their own provenance boundary. No numerical payload in this report is the sole record of a result.

References

[1] Nicola Marzari and David Vanderbilt. Maximally localized generalized wannier functions for composite energy bands. *Physical Review B*, 56(20):12847–12865, 1997.

[2] Ivo Souza, Nicola Marzari, and David Vanderbilt. Maximally localized wannier functions for entangled energy bands. *Physical Review B*, 65(3):035109, 2001.

[3] Arash A. Mostofi, Jonathan R. Yates, Giovanni Pizzi, Young-Su Lee, Ivo Souza, David Vanderbilt, and Nicola Marzari. An updated version of wannier90: A tool for obtaining maximally-localised wannier functions. *Computer Physics Communications*, 185(8):2309–2310, 2014.

[4] Jonathan R. Yates, Xinjie Wang, David Vanderbilt, and Ivo Souza. Spectral and fermi surface properties from wannier interpolation. *Physical Review B*, 75(19):195121, 2007.

[5] J. M. Luttinger and W. Kohn. Motion of electrons and holes in perturbed periodic fields. *Physical Review*, 97(4):869–883, 1955.